MSG No. 73.551 $Ib'c'a$ [Type III, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3]`
2	$[0, \frac{1}{2}, 0]$	`[2,4]`
3	$[0, 0, \frac{1}{2}]$	`[5,7]`
4	$[\frac{1}{2}, 0, 0]$	`[6,8]`
5	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,11]`
6	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[10,12]`
7	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[13,15]`
8	$[0, \frac{1}{2}, \frac{1}{2}]$	`[14,16]`

Table 2: Wyckoff site: `8b`, site symmetry: `-1`

No.	position	mapping
1	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,11]`
2	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,12]`
3	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,9]`
4	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,10]`
5	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,15]`
6	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[6,16]`
7	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7,13]`
8	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[8,14]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'..`

No.	position	mapping
1	$[x, 0, \frac{1}{4}]$	`[1,5]`
2	$[-x, \frac{1}{2}, \frac{1}{4}]$	`[2,14]`
3	$[-x, 0, \frac{3}{4}]$	`[3,7]`
4	$[x, \frac{1}{2}, \frac{3}{4}]$	`[4,16]`
5	$[\frac{1}{2} - x, 0, \frac{3}{4}]$	`[6,10]`
6	$[x + \frac{1}{2}, 0, \frac{1}{4}]$	`[8,12]`
7	$[x + \frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[9,13]`
8	$[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{4}]$	`[11,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2'.`

No.	position	mapping
1	$[\frac{1}{4}, y, 0]$	`[1,6]`
2	$[\frac{3}{4}, \frac{1}{2} - y, 0]$	`[2,13]`
3	$[\frac{3}{4}, -y, 0]$	`[3,8]`
4	$[\frac{1}{4}, y + \frac{1}{2}, 0]$	`[4,15]`
5	$[\frac{1}{4}, -y, \frac{1}{2}]$	`[5,10]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	[7,12]
7	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,14]
8	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}\right]$	[11,16]

Table 5: Wyckoff site: 8e, site symmetry: ..2

No.	position	mapping
1	$\left[0, \frac{1}{4}, z\right]$	[1,2]
2	$\left[0, \frac{3}{4}, -z\right]$	[3,4]
3	$\left[0, \frac{3}{4}, \frac{1}{2}-z\right]$	[5,14]
4	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	[6,13]
5	$\left[0, \frac{1}{4}, z+\frac{1}{2}\right]$	[7,16]
6	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	[8,15]
7	$\left[\frac{1}{2}, \frac{3}{4}, z+\frac{1}{2}\right]$	[9,10]
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	[11,12]

Table 6: Wyckoff site: 16f, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[-x, \frac{1}{2}-y, z\right]$	[2]
3	$[-x, -y, -z]$	[3]
4	$\left[x, y+\frac{1}{2}, -z\right]$	[4]
5	$\left[x, -y, \frac{1}{2}-z\right]$	[5]
6	$\left[\frac{1}{2}-x, y, -z\right]$	[6]
7	$\left[-x, y, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2}, -y, z\right]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	[10]
11	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[11]
12	$\left[x+\frac{1}{2}, y, \frac{1}{2}-z\right]$	[12]
13	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[13]
14	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z\right]$	[15]
16	$\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[16]